



Dear Editor:

Uveal melanoma is the most common primary intraocular malignancy reported in adults, accounting for 70% of all intraocular tumors.¹ In the United States, about 6 cases per million population are believed to be diagnosed annually, accounting for 1600 to 2000 new cases.^{1,2} Enucleation has been the standard treatment for eyes containing uveal melanoma in the past, and despite treatment, the 5-year prognosis is poor, with mortality rates ranging from 25% to 50%, depending on tumor size.³ The Collaborative Ocular Melanoma Study results indicate that for medium-size choroidal melanoma, enucleation and iodine 125 brachytherapy offered similar survival rates.⁴ The demographics of this intraocular malignancy have been studied extensively in the Caucasian population from Western countries.¹⁻⁴ Because of its rarity in non-Caucasian patients, studies of this tumor seldom are reported for the non-Caucasian population, and only a few reports of uveal melanoma have surfaced from developing countries comprising mainly non-Caucasian populations.⁵ The purpose of this study is to present the demographics and occurrence of uveal melanoma from the non-Caucasian Arab population of Saudi Arabia.

A medical record review of clinical history, imaging studies, surgical procedures, and treatment outcomes of patients treated for uveal melanoma at King Khaled Eye Specialist Hospital between 1983 and 2005 was performed. Among the 60 patients (37 male; average age, 51 years), decreased visual acuity (VA) was the main presenting complaint of 47 (78%) patients, followed by pain in 12 (20%). Duration of the presenting symptoms was within 2 months for 19 (32%) patients, between 3 and 5 months for 15 (25%), and >6 months in 24 (40%). The apical height of the tumor was >5 mm in 50 (83%) of the eyes, and the basal dimension was >15 mm in 24 (40%) (Table 1 [available at <http://aaojournal.org>]). At presentation, VAs were >20/100 in 13 (25%) eyes and <20/200 in 40 (75%) eyes. The tumor involved the macula in 4 eyes, the optic disc in 4, and within 1 to 3 mm of the optic disc in 4. Subretinal fluid and associated retinal detachment was present in 48 (80%) eyes. In 59 (98%) eyes, the diagnosis of uveal melanoma was made on clinical and/or echographic findings. Six (10%) eyes were treated with episcleral radiation plaque therapy, and 2 had endoresection, with simultaneous episcleral radioactive plaque therapy in 1. Enucleation was performed in 47 (73%) eyes and partial exenteration in 3 (5%) orbits due to evidence of extraocular tumor extension. Six (10%) patients underwent additional external beam radiation treatment due to the presence of extraocular tumor extension. The diagnosis of uveal melanoma was confirmed by histopathology in 49 (98%) eyes. One patient presented with no light perception and clinical features of endophthalmitis. No

intraocular mass was detected by ultrasound, and the diagnosis of choroidal melanoma was made only after enucleation of the eye. On histopathology, 31 (63%) tumors were found to have spindle cell-type uveal melanoma, and 18 (37%) tumors had a mixed or epithelioid cell type. Metastatic uveal melanoma was detected after the initial treatment in 7 (12%) patients, ranging from 6 to 64 months after the initial treatment of their uveal melanoma. The site of metastasis included liver (4), lids (2), scalp (1), and optic chiasm (1) (Table 2 [available at <http://aaojournal.org>]). Average follow-up was 22.5 months (range, 2 months–15 years).

Fewer than 3 cases of uveal melanoma per year from a major eye care referral center in the Middle East confirms previous observations that uveal melanoma primarily affects Caucasians.¹⁻⁴ Although demographics of uveal melanoma from different ethnic groups have been reported from several regions of the world, to our knowledge this is the first report describing uveal melanoma demographics from an Arab population. Currently, there is no well-developed registry for newly diagnosed cases of uveal melanoma in Saudi Arabia. Although there were 567 new cases of retinoblastoma diagnosed, there were only 60 cases of uveal melanoma diagnosed and treated during the same period. This relatively low figure gives the impression of a very low incidence of uveal melanoma in the Arab population. Further studies are required to estimate the true incidence of uveal melanoma in an Arab population. Early diagnosis and treatment may improve the outcome in most cases of uveal melanoma.

ADEL AL-SUHAIBANI, MD
IMTIAZ A. CHAUDHRY, MD, PhD
ESSAM AL-HARTHI, MD
EMAD B. ABBOD, MD
Riyadh, Saudi Arabia

References

1. Jemal A, Thomas A, Murray T, Thun M. Cancer statistics, 2002. *CA Cancer J Clin* 2002;52:23–47.
2. Singh AD, Topham A. Survival rates with uveal melanoma in the United States: 1973–1997. *Ophthalmology* 2003;110:962–5.
3. Diener-West M, Hawkins BS, Markowitz JA, Schachat AP. A review of mortality from choroidal melanoma. II. A meta-analysis of 5-year mortality rates following enucleation, 1966 through 1988. *Arch Ophthalmol* 1992;110:245–50.
4. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: initial mortality findings. COMS report no. 18. *Arch Ophthalmol* 2001;119:969–82.
5. Kuo PK, Puliafito CA, Wang KM, et al. Uveal melanoma in China. *Int Ophthalmol Clin* 1982;22(3):57–71.